CHAPTER 13
barrier options

The aim of this Chapter...

...is to describe and classify barrier options, to show how they can easily be put into a partial differential equation framework for later solution by numerical methods. Such a framework is ideal for pricing barrier options with complex modern volatility models.

In this Chapter...

- the different types of barrier options
- how to price many barrier contracts in the partial differential equation framework
- some of the practical problems with the pricing and hedging of barriers

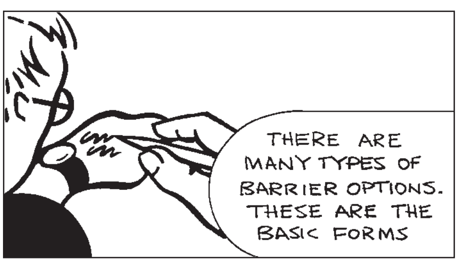
13.1 INTRODUCTION

I mentioned barrier options briefly in an earlier chapter. In this chapter we study them in detail, from both a theoretical and a practical perspective. **Barrier options** are path-dependent options. They have a payoff that is dependent on the realized asset path via its level; certain aspects of the contract are triggered if the asset price becomes too high or too low. For example, an up-and-out call option pays off the usual $\max(S - E, 0)$ at expiry unless at any time previously the underlying asset has traded at a value S_u or higher. In this example, if the asset reaches this level (from below, obviously) then it is said to 'knock out,' becoming worthless. Apart from 'out' options like this, there are also 'in' options which only receive a payoff if a level is reached, otherwise they expire worthless.

Barrier options are popular for a number of reasons. Perhaps the purchaser uses them to hedge very specific cashflows with similar properties. Usually, the purchaser has very precise views about the direction of the market. If he wants the payoff from a call option but does not want to pay for all the upside potential, believing that the upward movement of the underlying will be limited prior to expiry, then he may choose to buy an up-and-out call. It will be cheaper than a similar vanilla call, since the upside is severely limited. If he is right and the barrier is not triggered he gets the payoff he wanted. The closer that the barrier is to the current asset price then the greater the likelihood of the option being knocked out, and thus the cheaper the contract.

Conversely, an 'in' option will be bought by someone who believes that the barrier level will be realized. Again, the option is cheaper than the equivalent vanilla option.

13.2 DIFFERENT TYPES OF BARRIER OPTIONS



There are two main types of barrier option:

- The **out option**, which only pays off if a level is *not* reached. If the barrier is reached then the option is said to have **knocked out**.
- The **in option**, which pays off as long as a level is reached before expiry. If the barrier is reached then the option is said to have **knocked in**.

Then we further characterize the barrier option by the position of the barrier relative to the initial value of the underlying:

- If the barrier is above the initial asset value, we have an **up** option.
- If the barrier is below the initial asset value, we have a **down** option.

Finally, we describe the payoff received at expiry:

- The payoffs are all the usual suspects, call, put, binary, etc.

The above classifies the commonest barrier options. In all of these contracts the position of the barrier could be time dependent. The level may begin at one level and then rise, say. Usually the level is a piecewise-constant function of time.

Preliminary and Indicative
For Discussion Purposes Only

USD/MXN Double Knock-Out Note

Principal Amount	USD 10,000,000
Issuer	XXXX
Maturity	6 months from Trade Date
Issue Price	100%
Coupon	If the USD/MXN spot exchange rate trades above the Upper Barrier orbelow the Lower Barrier at any time during the term of the Note: Zero Otherwise: $400\% \times \max\left(0, \frac{8.2500\text{-FX}}{\text{FX}}\right)$ where FX isthe USD/MXN spot exchage rate at Maturity
Redemption Amount	100%
Upper Barrier Level	8.2500
Lower Barrier Level	7.4500

Figure 13.1 Term sheet for a USD/MXN double knock-out note.

Another style of barrier option is the **double barrier**. Here there is both an upper and a lower barrier, the first above and the second below the current asset price. In a double 'out' option the contract becomes worthless if *either* of the barriers is reached. In a double 'in' option one of the barriers must be reached before expiry, otherwise the option expires worthless. Other possibilities can be imagined, one barrier is an 'in' and the other an 'out,' at expiry the contract could have either an 'in' or an 'out' payoff.

Sometimes a **rebate** is paid if the barrier level is reached. This is often the case for 'out' barriers in which case the rebate can be thought of as cushioning the blow of losing the rest of the payoff. The rebate may be paid as soon as the barrier is triggered or not until expiry.

In Figure 13.1 is shown the term sheet for a double knock-out option on the Mexican peso, US dollar exchange rate. The upper barrier is set at 8.25 and the lower barrier at 7.45. If the exchange rate trades inside this range until expiry then there is a payment. This is a very vanilla example of a barrier contract.

13.3 PRICING METHODOLOGIES

13.3.1 Monte Carlo simulation

Pricing via Monte Carlo simulation is simple in principle:

- The value of an option is the present value of the expected payoff under a risk-neutral random walk.

The pricing algorithm:

1. Simulate the risk-neutral random walk starting at today's value of the asset over the required time horizon. This gives one realization of the underlying price path.
2. For this realization calculate the option payoff.
3. Perform many more such realizations over the time horizon.
4. Calculate the average payoff over all realizations.
5. Take the present value of this average, this is the option value.

Advantages of Monte Carlo pricing

- It is easy to code.
- It is hard to make mistakes in the coding.

Disadvantages of Monte Carlo pricing

- More work is needed to get the greeks.
- It can be slow since tens of thousands of simulations are needed to get an accurate answer.

13.3.2 Partial differential equations

Barrier options are path dependent. Their payoff, and therefore value, depends on the path taken by the asset up to expiry.

- Yet that dependence is weak. We only have to know whether or not the barrier has been triggered, we do not need any other information about the path.

13.4 PRICING BARRIERS IN THE PARTIAL DIFFERENTIAL EQUATION FRAMEWORK

I use $V(S, t)$ to denote the value of the barrier contract *before the barrier has been triggered*. This value still satisfies the Black–Scholes equation

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

The details of the barrier feature come in through the specification of the boundary conditions.

Time Out...

The framework

Partial differential equations are the natural framework for pricing barrier options. When combined with the kind of numerical method described in Chapter 28 the solution is relatively straightforward. Indeed, in many cases it is easier to solve a barrier option numerically than a vanilla option.

Once you are working in this framework it is very easy to incorporate modern volatility models. As hinted at later in this chapter, barrier options can be very sensitive to the assumptions made about volatility and very rarely are such options priced using the assumption of constant volatility. In such a situation it's nice to have a framework that can be easily adapted to sophisticated volatility models. This is that framework.

However, in the following I will also give a few pointers to valuing via trees.

13.4.1 'Out' barriers

If the underlying asset reaches the barrier in an 'out' barrier option then the contract becomes worthless. This leads to the boundary condition

$$V(S_u, t) = 0 \quad \text{for } t < T,$$

for an up-barrier option with the barrier level at $S = S_u$. We must solve the Black–Scholes equation for $0 \le S \le S_u$ with this condition on $S = S_u$ and a final condition corresponding to the payoff received if the barrier is not triggered. For a call option we would have

$$V(S, T) = \max(S - E, 0).$$

If we have a down-and-out option with a barrier at S_d then we solve for $S_d \le S < \infty$ with

$$V(S_d, t) = 0,$$

and the relevant final condition at expiry.

The boundary conditions are easily changed to accommodate rebates. If a rebate of R is paid when the barrier is hit then

$$V(S_d, t) = R.$$

Time Out...

In terms of the binomial model?

To price out barrier options in the binomial framework you have to specify on which nodes the option becomes valueless. The option value is then set to zero on these nodes, and the usual algorithm is used for all other nodes. (Of course, there is no need to find option values beyond the barrier, so the tree is usually smaller than for a vanilla option.)

13.4.2 'In' barriers

An 'in' option only has a payoff if the barrier is triggered. If the barrier is not triggered then the option expires worthless

$$V(S, T) = 0.$$

The value in the option is in the potential to hit the barrier. If the option is an up-and-in contract then on the upper barrier the contract must have the same value as a vanilla contract:

$$V(S_u, t) = \text{value of vanilla contract, a function of } t.$$

Using the notation $V_v(S, t)$ for value of the equivalent vanilla contract (a vanilla call, if we have an up-and-in call option) then we must have

$$V(S_u, t) = V_v(S_u, t) \quad \text{for } t < T.$$

A similar boundary condition holds for a down-and-in option.

The contract we receive when the barrier is triggered is a derivative itself, and therefore the 'in' option is a second-order contract.

In solving for the value of an 'in' option completely numerically we must solve for the value of the vanilla option first, before solving for the value of the barrier option. The solution therefore takes roughly twice as long as the solution of the 'out' option.[1]

When volatility is constant we can solve for the theoretical price of many types of barrier contract. Some examples are given at the end of the chapter.

[1] And, of course, the vanilla option must be solved for $0 \le S < \infty$.

Time Out...

In terms of the binomial model?

The ‘in’-barrier option can be thought of as a second-order contract. So you would need one tree for pricing the underlying option and the results of this will be passed to the ‘in’-barrier tree. To make things simple, make sure that the two trees have the same structure.

13.5 EXAMPLES

Down-and-out call option As the first example, consider the down-and-out call option with barrier level S_d below the strike price E. The value of this option is shown as a function of S in Figure 13.2.

Down-and-in call option In the absence of any rebates the relationship between an ‘in’-barrier option and an ‘out’-barrier option (with same payoff and same barrier level) is

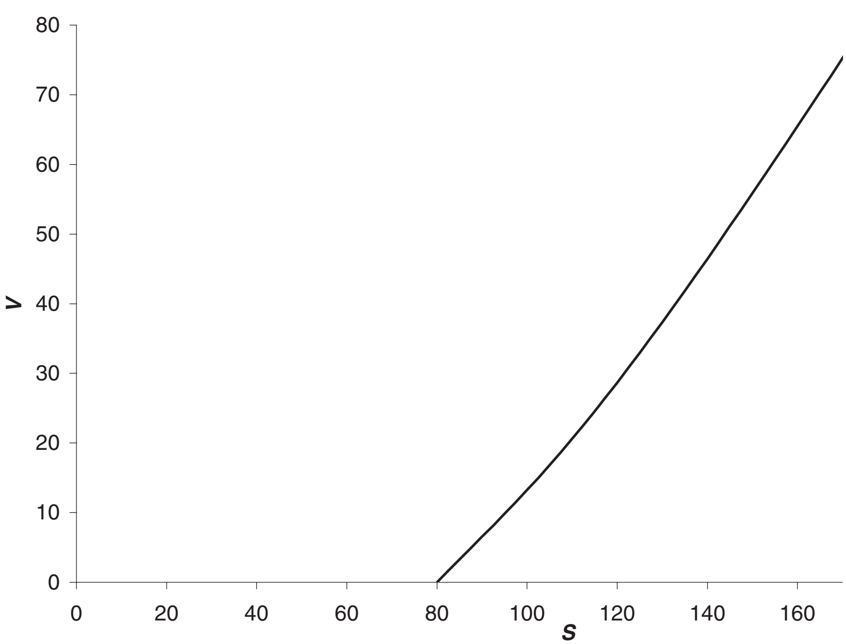


Figure 13.2 Value of a down-and-out call option.

very simple:

$$\text{in} + \text{out} = \text{vanilla}.$$

If the 'in' barrier is triggered then so is the 'out' barrier, so whether or not the barrier is triggered we still get the vanilla payoff at expiry.

The value of this option is shown as a function of S in Figure 13.3. Also shown is the value of the vanilla call. Note that the two values coincide at the barrier.

Up-and-out call option The barrier S_u for an up-and-out call option must be above the strike price E (otherwise the option would be valueless).

The value of this option is shown as a function of S in Figure 13.4. In Figure 13.5 is shown the delta.

Figure 13.6 shows the Bloomberg barrier option calculator and Figure 13.7 shows the option *profit/loss* against asset price.

13.5.1 Some more examples

The following figures are all taken from Bloomberg, who use the formulæ below, for the pricing.

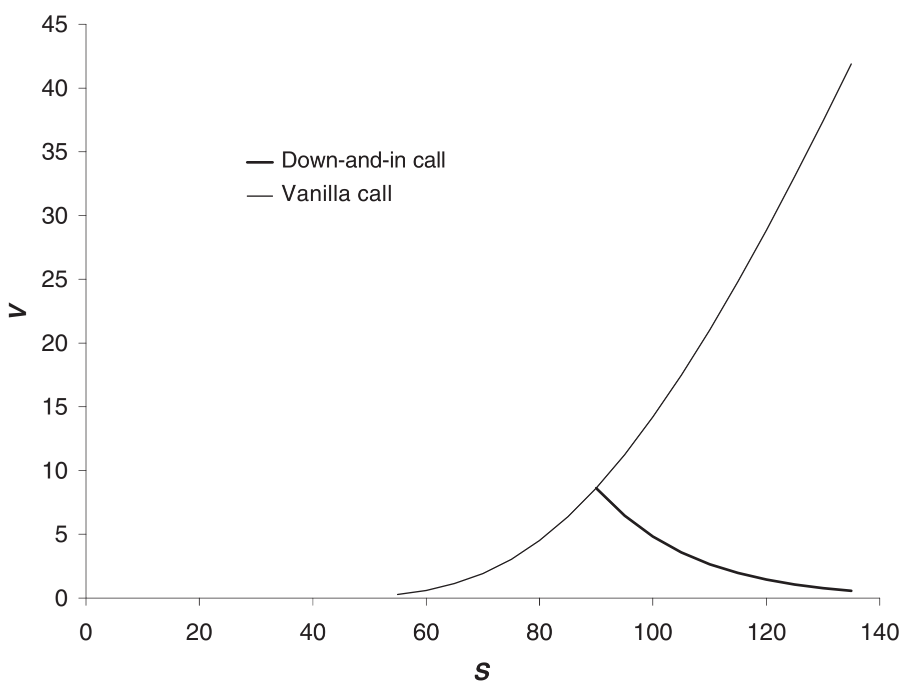


Figure 13.3 Value of a down-and-in call option.

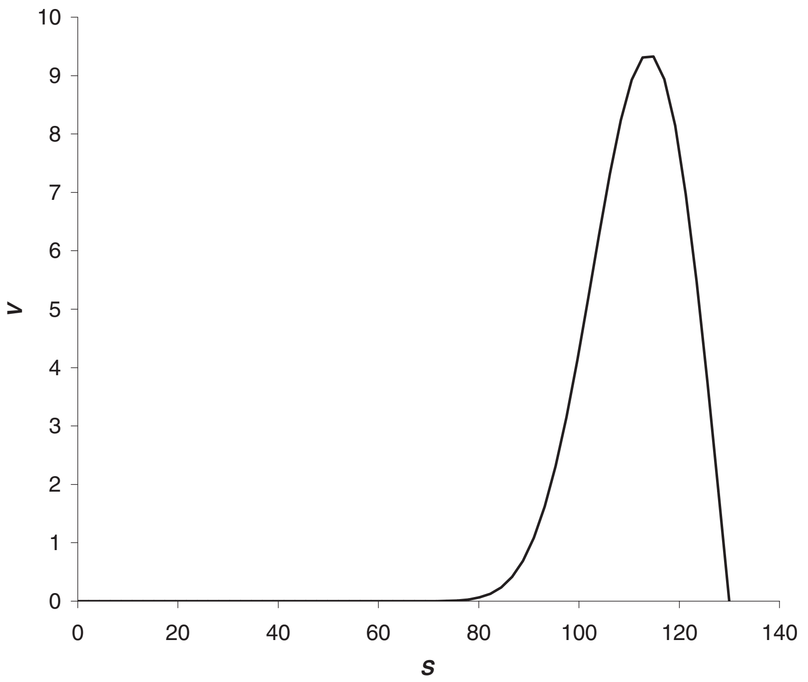


Figure 13.4 Value of an up-and-out call option.

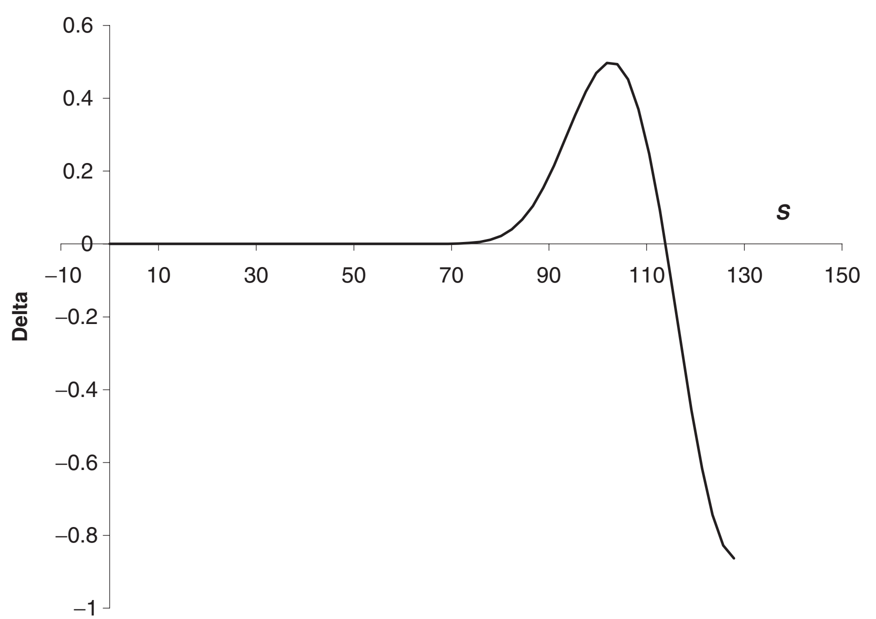


Figure 13.5 Delta of an up-and-out call option.

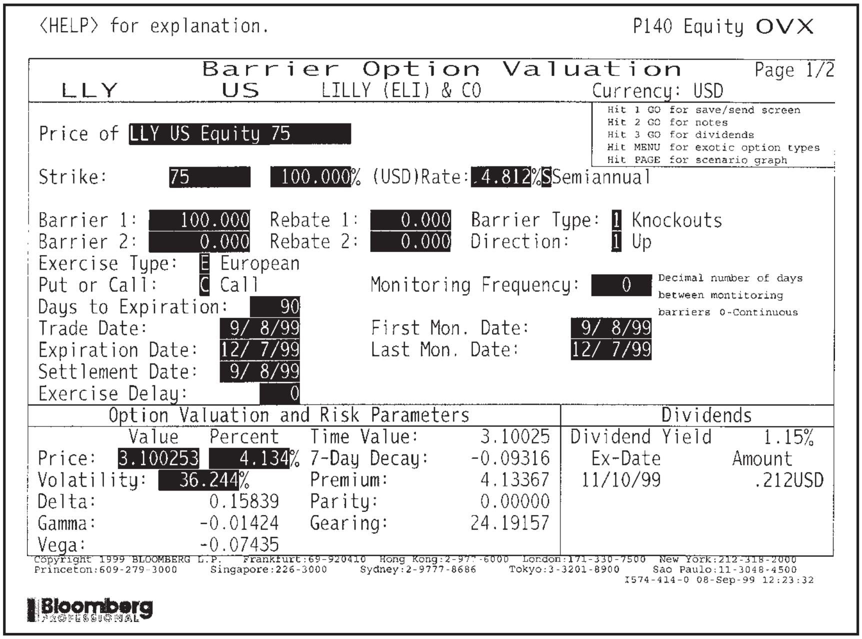


Figure 13.6 An up-and-out call again. Source: Bloomberg L.P.

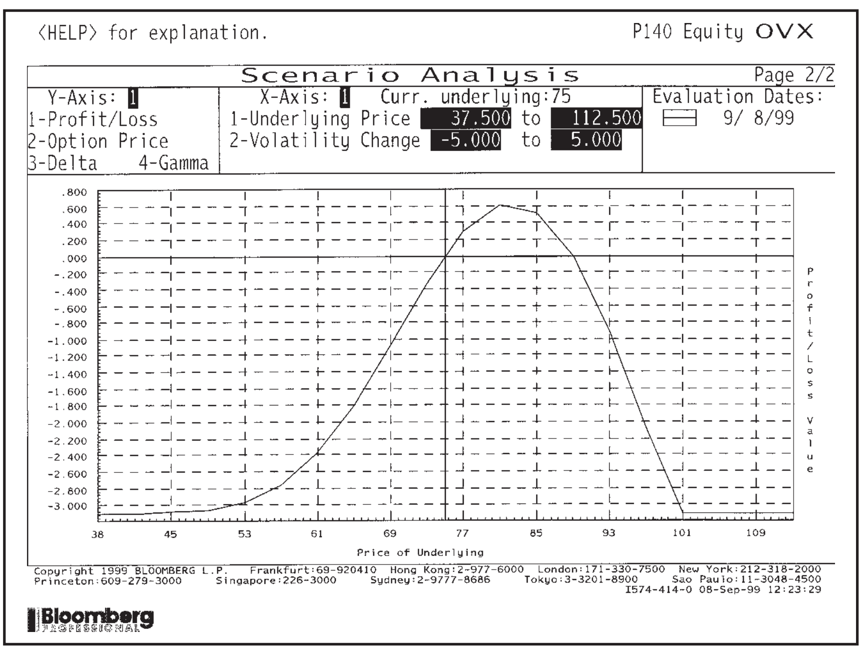
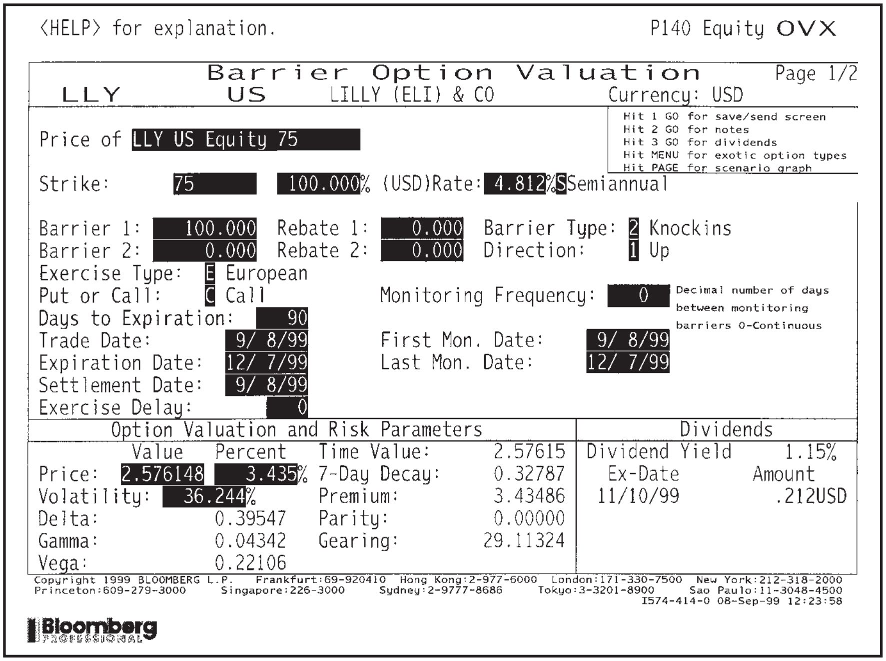


Figure 13.7 Profit/loss for an up-and-out call. Source: Bloomberg L.P.



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Barrier Option Valuation Page 1/2

LLY US LILLY (ELI) & CO Currency: USD

Hit 1 GO for save/send screen
Hit 2 GO for notes
Hit 3 GO for dividends
Hit MENU for exotic option types
Hit PAGE for scenario graph

Price of LLY US Equity 75

Strike: 75 100.000% (USD)Rate: 4.812% S Semiannual

Barrier 1: 100.000 Rebate 1: 0.000 Barrier Type: 2 Knockins
Barrier 2: 0.000 Rebate 2: 0.000 Direction: 1 Up
Exercise Type: E European
Put or Call: C Call Monitoring Frequency: 0 Decimal number of days between montitoring barriers 0-Continuous
Days to Expiration: 90
Trade Date: 9/ 8/99 First Mon. Date: 9/ 8/99
Expiration Date: 12/ 7/99 Last Mon. Date: 12/ 7/99
Settlement Date: 9/ 8/99
Exercise Delay: 0

Option Valuation and Risk Parameters					Dividends	
	Value	Percent	Time Value:	2.57615	Dividend Yield	1.15%
Price:	2.576148	3.435%	7-Day Decay:	0.32787	Ex-Date	Amount
Volatility:	36.244%		Premium:	3.43486	11/10/99	.212USD
Delta:		0.39547	Parity:	0.00000		
Gamma:		0.04342	Gearing:	29.11324		
Vega:		0.22106				

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Figure 13.8 Calculator for an up-and-in call. Source: Bloomberg L.P.

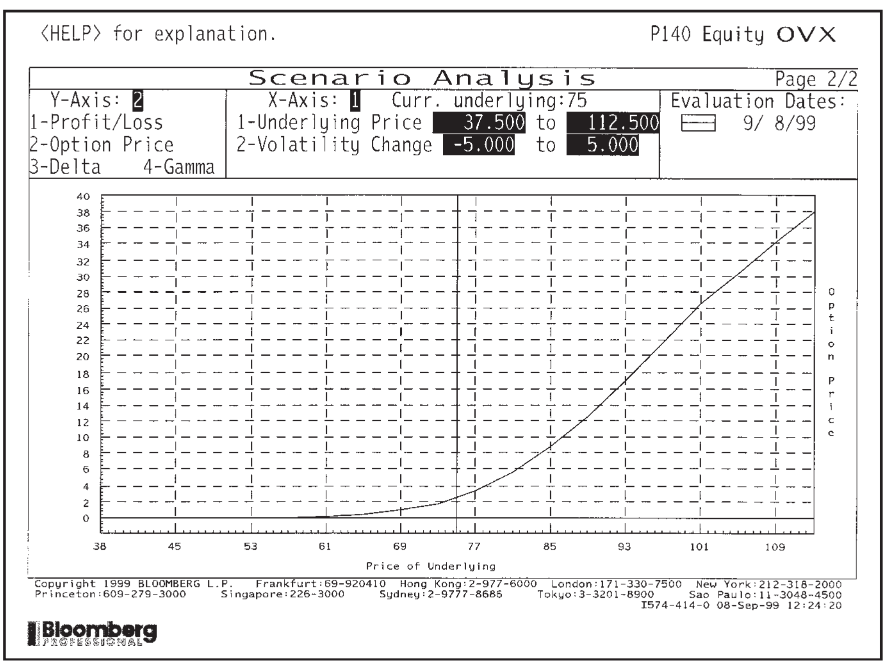

Figure 13.9 Value of an up-and-in call. Source: Bloomberg L.P.

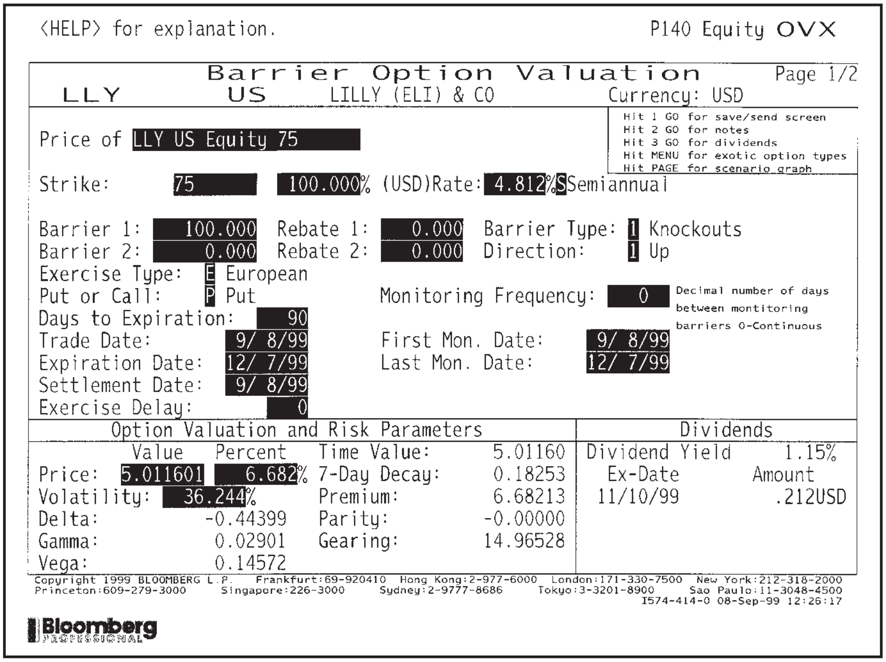

Figure 13.10 Calculator for an up-and-out put. Source: Bloomberg L.P.

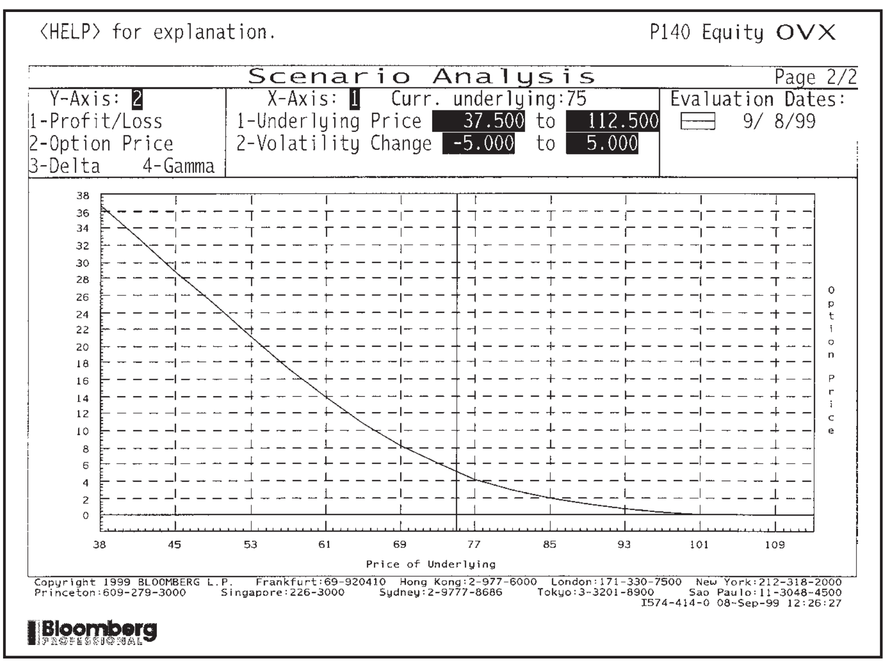

Figure 13.11 Value of an up-and-out put. Source: Bloomberg L.P.

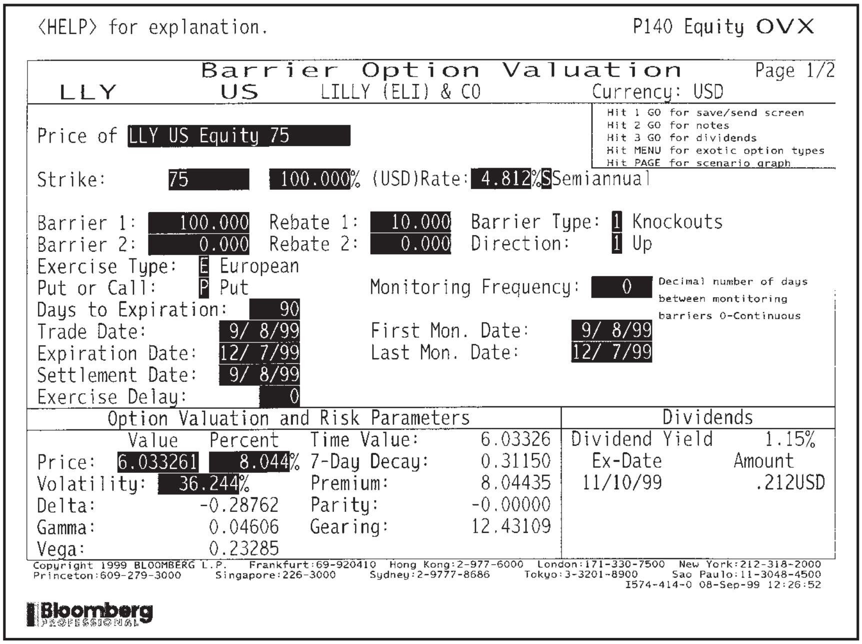

Figure 13.12 Calculator for an up-and-out put with a rebate on the upper barrier. Source: Bloomberg L.P.

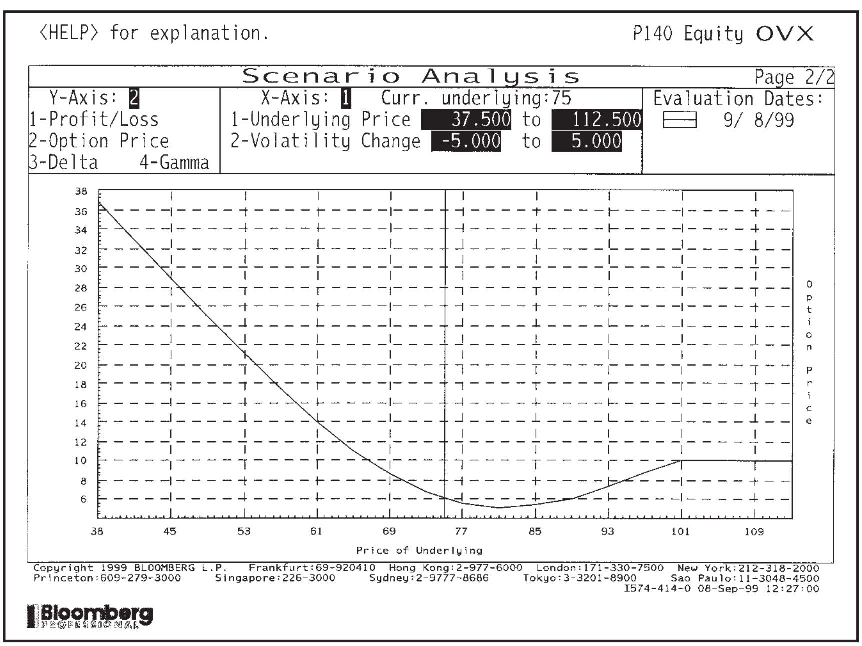

Figure 13.13 Value of an up-and-out put with a rebate on the upper barrier. Source: Bloomberg L.P.

13.6 OTHER FEATURES IN BARRIER-STYLE OPTIONS

Not so long ago barrier options were exotic, the market for them was small and few people were comfortable pricing them. Nowadays they are heavily traded and it is only the contracts with more unusual features that can rightly be called exotic. Some of these features are described below.

13.6.1 Early exercise

It is possible to have American-style early exercise. The contract must specify what the payoff is if the contract is exercised before expiry. As always, early exercise is a simple constraint on the value of the option.

In Figure 13.14 is the term sheet for a knock-out installment premium option on the US dollar, Japanese yen exchange rate. This knocks out if the exchange rate ever goes above 140. If the option expires without ever hitting this level there is a vanilla call payoff. I mention this contract in the section on early exercise because it has a similar feature. To keep the contract alive the holder must pay in instalments, every month another payment is due. The question is when to stop paying the instalments. This can be done optimally.

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USD/JPY KO Installment-Premium Option

Notional Amount	USD 50,000,000
Option Type	133.25 (ATMS) USD Put/JPY Call with KO and Installment Premium
Maturity	6 months from Trade Date
Knockout Mechanism	If, at any time from Trade Date to Maturity, the USD/JPY spot rate trades in the interbank market at or above JPY 140.00per USD, the option will automatically be cancelled, with no further rights or obligations arising for the parties thereto.
Upfront Premium	JPY 1.50 per USD
Installments	JPY 1.50 per USD,payable monthly from Trade Date (5 installments)
Installment Mechanism	As long as the installments continue to be paid, the option will be kept alive, but the Counterparty has the right to cease paying the installments and to thereby let the option becancelled at anytime.
Spot Reference	JPY 133.25 per USD

This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 13.14 Term sheet for a USD/JPY knock-out installment premium option.

13.6.2 Repeated hitting of the barrier

The double barrier that we have seen above can be made more complicated. Instead of only requiring one hit of either barrier we could insist that *both* barriers are hit before the barrier is triggered.

This contract is easy to value. Observe that the first time that one of the barriers is hit the contract becomes a vanilla barrier option. Thus on the two barriers we solve the Black–Scholes equation with boundary conditions that our double barrier value is equal to an up-barrier option on the lower barrier and a down-barrier option on the upper barrier.

13.6.3 Resetting of barrier

Another type of barrier contract that can be priced by the same two- (or more) step procedure as 'in' barriers is the reset barrier. When the barrier is hit the contract turns into another barrier option with a different barrier level. The contract may be time dependent in the sense that if the barrier is hit before a certain time we get a new barrier option, if it is hit after a certain time we get the vanilla payoff.

Related to these contracts are the **roll-up** and **roll-down options**. These begin life as vanilla options, but if the asset reaches some predefined level they become a barrier option. For example, with a roll-up put if the roll-up strike level is reached the contract becomes an up-and-out put with the roll-up strike being the strike of the barrier put. The barrier level will then be at a prespecified level.

13.6.4 Outside barrier options

Outside or **rainbow barrier options** have payoffs or a trigger feature that depends on a second underlying. Thus the barrier might be triggered by one asset, with the payoff depending on the other. These products are clearly multi-factor contracts.

13.6.5 Soft barriers

The **soft barrier option** allows the contract to be gradually knocked in or out. The contract specifies two levels, an upper and a lower. In the knock-out option a proportion of the contract is knocked out depending on the distance that the asset has reached between the two barriers. For example, suppose that the option is an up and out with a soft barrier range of 100 to 120. If the maximum asset value reached before expiry is 105 then 5/20 or 25% of the payoff is lost.

13.6.6 Parisian options

Parisian options have barriers that are triggered only if the underlying has been beyond the barrier level for more than a specified time. This additional feature reduces the possibility of manipulation of the trigger event and makes the dynamic hedging easier. However, this new feature also increases the dimensionality of the problem.

13.7 MARKET PRACTICE: WHAT VOLATILITY SHOULD I USE?

Practitioners do not price contracts using a single, constant volatility. Let us see some of the pitfalls with this, and then see what practitioners do.

In Figure 13.15 we see a plot of the value of an up-and-out call option using three different volatilities, 15%, 20% and 25%. I have chosen three very different values to make a point. If we are unsure about the value of the volatility (as we surely are) then which value do we use to price the contract? Observe that at approximately $S = 100$ the option value seems to be insensitive to the volatility, the vega is zero. If S is greater than this value perhaps we should only sell the contract for a volatility of 15% to be on the safe side. If S is less than this, perhaps we should sell the contract for 25%, again to play it safe. Now ask the question: Do I believe that volatility will be one of 15%, 20% or 25%, and will be fixed at that level? Or do I believe that volatility could move around between 15% and 25%? Clearly the latter is closer to the truth. But the measurement of vega and the plots in Figure 13.15 assume that volatility is fixed until expiry. If we are concerned with playing it safe we should assume that the behavior of volatility will be that which gives us the lowest value if we are buying the contract. The worst outcome for volatility is for it to be low below the strike price and high around the barrier. Financially, this means that if we are near the strike we get a small payoff, but if we are near the barrier we are likely to hit it. Mathematically, the 'worst' choice of volatility path depends on the sign of the gamma at each point. If gamma is positive then low volatility is bad,

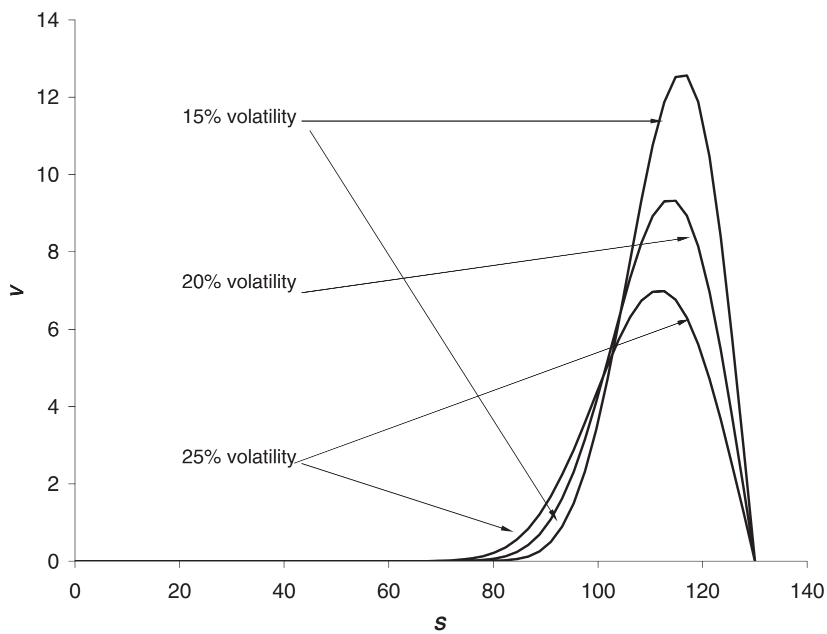
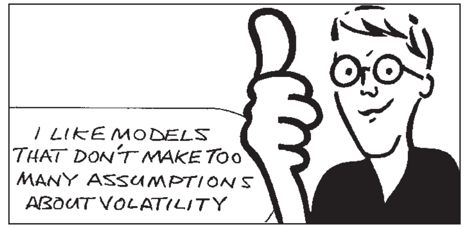


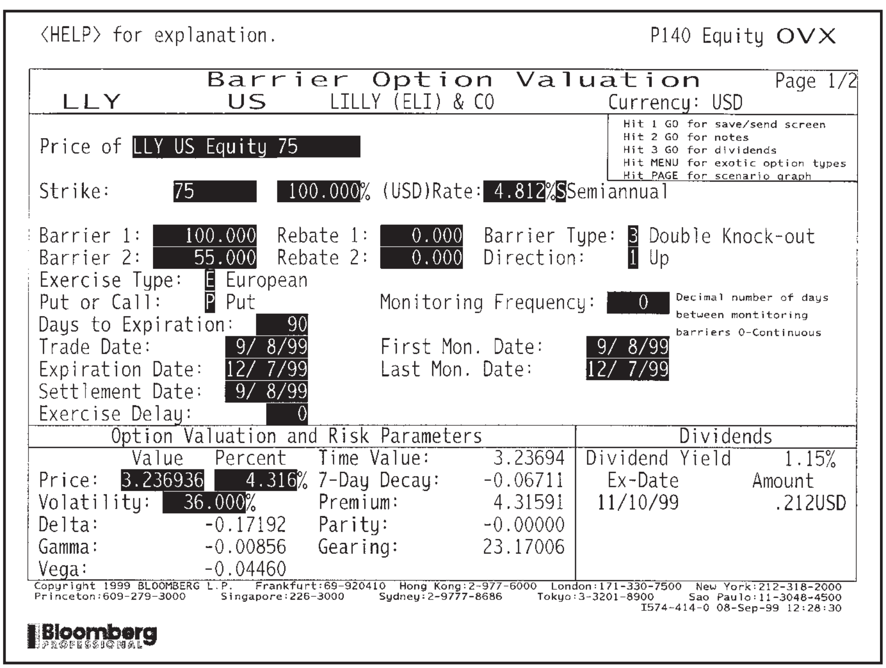
Figure 13.15 Theoretical up-and-out call price with three different volatilities. Source: Bloomberg L.P.



if gamma is negative then high volatility is bad. When the gamma is not single signed, the measurement of vega can be meaningless. Barrier options with non-single-signed gamma include the up-and-out call, down-and-out put and many double-barrier options.

Figures 13.16 through 13.19 show the details of a double knockout put contract, its price versus the underlying, its gamma versus the underlying and its price versus volatility. This is a contract with a gamma that changes sign as can be seen from Figure 13.18. You must be very careful when pricing such a contract as to what volatility to use. Suppose you wanted to know the implied volatility for this contract when the price was 3.2, what value would you get? Refer to Figure 13.19.

To accommodate problems like this, practitioners have invented a number of 'patches.' One is to use two different volatilities in the option price. For example, one can calculate implied volatilities from vanilla options with the same strike, expiry and payoff as the barrier option and also from American-style one-touch options with the strike at the barrier level. The implied volatility from the vanilla option contains the market's estimate of the value of the payoff, but includes all the upside potential that the call has but which is irrelevant for the up-and-out option. The one-touch volatility, however, contains the market's view of the likelihood of the barrier level being reached. These two volatilities can be used to price an up-and-out call by observing that an 'out' option is the same as a vanilla minus an 'in' option. Use the vanilla volatility to price the vanilla call and the one-touch volatility to price the 'in' call.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Barrier Option Valuation Page 1/2
LLY US LILLY (ELI) & CO Currency: USD

Price of LLY US Equity 75

Hit 1 GO for save/send screen
Hit 2 GO for notes
Hit 3 GO for dividends
Hit MENU for exotic option types
Hit PAGE for scenario graph

Strike: 75 100.000% (USD)Rate: 4.812% S Semiannual

Barrier 1: 100.000 Rebate 1: 0.000 Barrier Type: 3 Double Knock-out
Barrier 2: 55.000 Rebate 2: 0.000 Direction: 1 Up
Exercise Type: E European
Put or Call: P Put Monitoring Frequency: 0 Decimal number of days between montitoring barriers 0-Continuous
Days to Expiration: 90
Trade Date: 9/ 8/99 First Mon. Date: 9/ 8/99
Expiration Date: 12/ 7/99 Last Mon. Date: 12/ 7/99
Settlement Date: 9/ 8/99
Exercise Delay: 0

Option Valuation and Risk Parameters	Value	Percent			Dividends	
			Time Value:	3.23694	Dividend Yield	1.15%
Price:	3.236936	4.316%	7-Day Decay:	-0.06711	Ex-Date	Amount
Volatility:	36.000%		Premium:	4.31591	11/10/99	.212USD
Delta:	-0.17192		Parity:	-0.00000		
Gamma:	-0.00856		Gearing:	23.17006		
Vega:	-0.04460					

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Princeton:609-279-3000 Singapore:226-3000 Sydney:2-9777-8686 Tokyo:3-3201-8900 Sao Paulo:11-3048-4500
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Figure 13.16 Details of a double knockout put. Source: Bloomberg L.P.

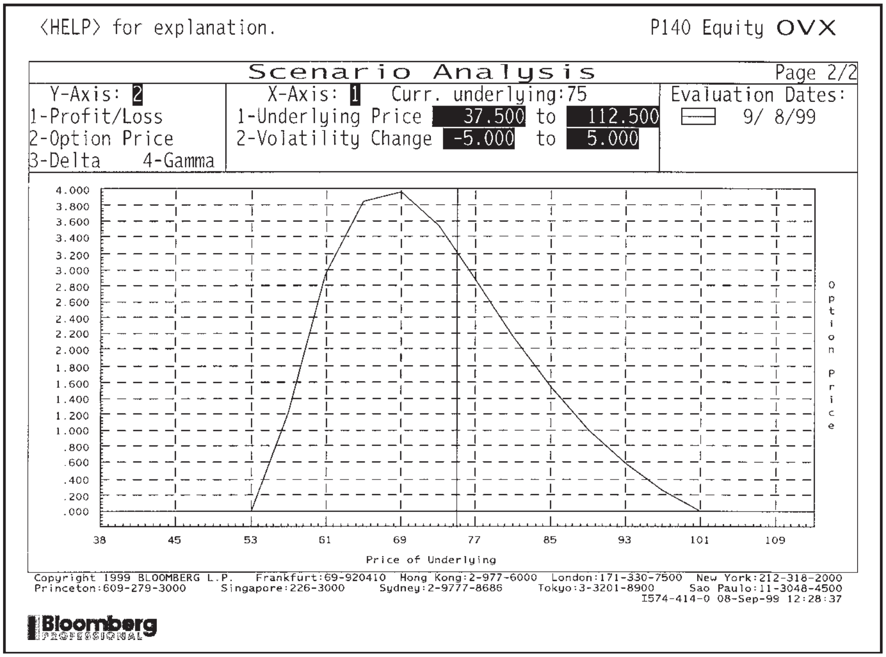

Figure 13.17 Price of the double knockout put. Source: Bloomberg L.P.

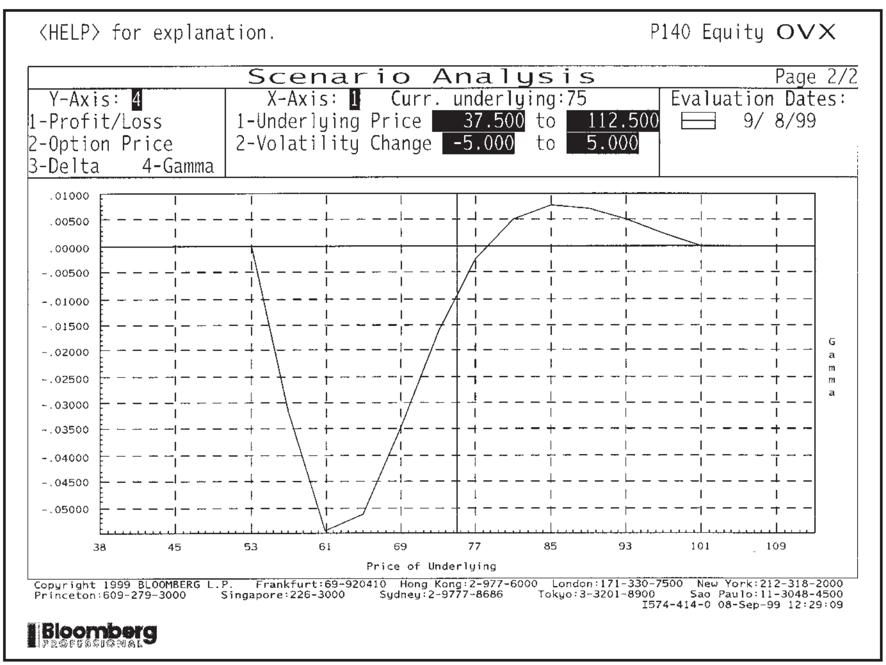

Figure 13.18 Gamma of the double knockout put. Source: Bloomberg L.P.

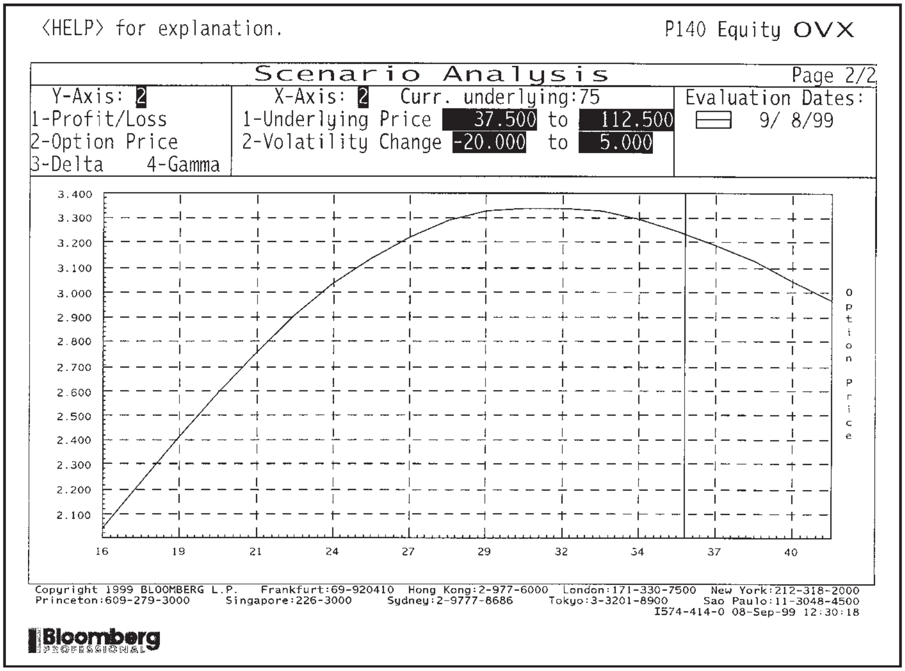

Figure 13.19 Option price versus volatility for the double knockout put. Source: Bloomberg L.P.

13.8 HEDGING BARRIER OPTIONS

Barrier options have discontinuous delta at the barrier. For a knock-out, the option value is continuous, decreasing approximately linearly towards the barrier then being zero beyond the barrier. This discontinuity in the delta means that the gamma is instantaneously infinite at the barrier. Delta hedging through the barrier is virtually impossible, and certainly very costly. This raises the issue of whether there are improvements on delta hedging for barrier options.

There have been a number of suggestions made for ways to *statically* hedge barrier options. These methods try to mimic as closely as possible the value of a barrier option with vanilla calls and puts, or with binary options. A very common practice for hedging a short up-and-out call is to buy a long call with the same strike and expiry. If the option does knock out then you are fortunate in being left with a long call position.

I now describe another simple but useful technique, based on the **reflection principle** and **put-call symmetry**. This technique only really works if the barrier and strike lie in the correct order, as we shall see. The method gives an approximate hedge only.

The simplest example of put-call symmetry is actually put-call parity. At all asset levels we have

$$V_C - V_P = S - Ee^{-r(T-t)},$$

where E is the strike of the two options, and C and P refer to call and put. Suppose we have a down-and-in call, how can we use this result? To make things simple for the moment, let's have the barrier and the strike at the same level. Now hedge our down-and-in call with a short position in a vanilla put with the same strike. If the barrier is reached we have a position worth

$$V_C - V_P.$$

The first term is from the down-and-in call and the second from the vanilla put. This is exactly the same as

$$S - Ee^{-r(T-t)} = E(1 - e^{-r(T-t)}),$$

because of put-call parity and since the barrier and the strike are the same. If the barrier is not touched then both options expire worthless. If the interest rate were zero then we would have a perfect hedge. If rates are non-zero what we are left with is a one-touch option with small and time-dependent value on the barrier. Although this leftover cashflow is non-zero, it is small, bounded and more manageable than the original cashflows.

Now suppose that the strike and the barrier are distinct. Let us continue with the down-and-in call, now with barrier below the strike. The static hedge is not much more complicated than the previous example. All we need to know is the relationship between the value of a call option with strike E when $S = S_d$ and a put option with strike S_d^2/E. It is easy to show from the formulæ for calls and puts that if interest rates are zero, the value of this call at $S = S_d$ is equal to a number E/S_d of the puts, valued at S_d. We would therefore hedge our down-and-in call with E/S_d puts struck at S_d^2/E. Note that the geometric average of the strike of the call and the strike of the put is the same as the barrier level, this is where the idea of 'reflection' comes in. The strike of the hedging put is at the reflection in the barrier of the call's strike. When rates are non-zero there is some error in this hedge, but again it is small and manageable, decreasing as we get closer to expiry. If the barrier is not touched then both options expire worthless (the strike of the put is below the barrier remember).

If the barrier level is above the strike, matters are more complicated since if the barrier is touched we get an in-the-money call. The reflection principle does not work because the put would also be in the money at expiry if the barrier is not touched.

13.8.1 Slippage costs

The delta of a barrier option is discontinuous at the barrier, whether it is an in or an out option. This presents a particular problem to do with **slippage** or **gapping**. Should the underlying move significantly as the barrier is triggered it is likely that it will not be possible to continuously hedge through the barrier. For example, if the contract is knocked out then one finds oneself with a $-\Delta$ holding of the underlying that should have been offloaded sooner. This can have a significant effect on the hedging costs.

It is not too difficult to allow for the *expected* slippage costs, and all that is required is a slight modification to the apparent barrier level.

At the barrier we hold $-\Delta$ of the underlying. The value of this position is $-\Delta X$, since $S = X$ is the barrier level. Suppose that the asset moves by a small fraction k before

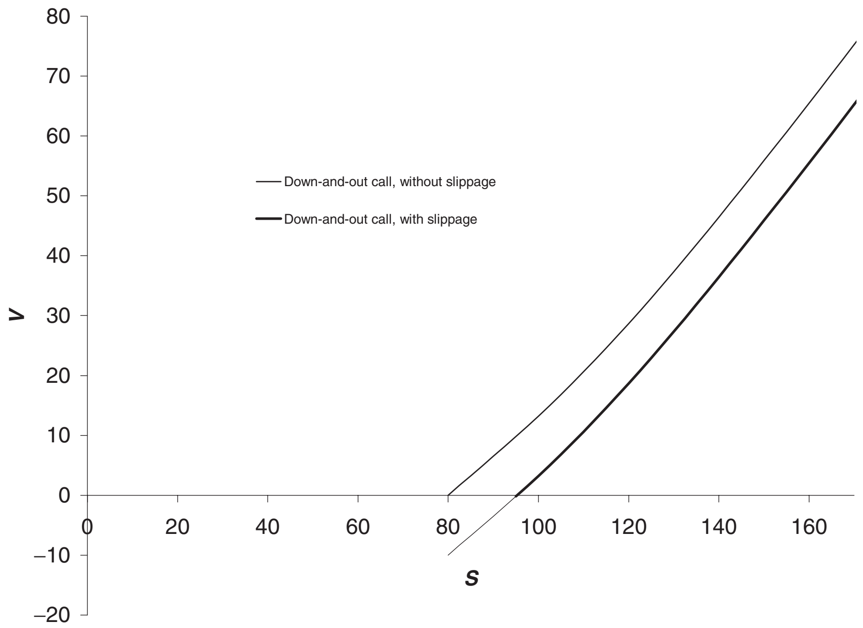

Figure 13.20 Incorporating slippage.

we can close out our asset position, or equivalently, that there is a transaction charge involved in closing. We thus lose

$$-k\Delta X$$

on the trigger event.

Now refer to Figure 13.20 where we'll look at the specific example of a down-and-out option. Because we lose $-k\Delta X$ we should use the boundary condition

$$V(X, t) = -k\Delta X.$$

After a little bit of Taylor series we find that this is approximately the same as

$$V((1 + k)X, t) = 0.$$

In other words, we should apply the boundary condition at a slightly higher value of S and so slightly reduce the option's value.

13.9 SUMMARY

In this chapter we have seen a description of many types of barrier option. We have seen how to put these contracts into the partial differential equation framework. Many of these contracts have simple pricing formulæ. Unfortunately, the extreme nature of these contracts make them very difficult to hedge in practice and in particular, they can be very sensitive to the volatility of the underlying. Worse still, if the gamma of the contract changes sign we cannot play safe by adding a spread to the volatility. Practitioners seem to be most comfortable statically hedging as much of the barrier contract as possible using traded vanillas.

All implmented on the CD

SOME FORMULÆ

In the following I use $N(\cdot)$ to denote the cumulative distribution function for a standardized Normal variable. The dividend yield on stocks or the foreign interest rate for FX are denoted by q. Also

$$a = \left(\frac{S_b}{S}\right)^{-1+\frac{2(r-q)}{\sigma^2}},$$

$$b = \left(\frac{S_b}{S}\right)^{1+\frac{2(r-q)}{\sigma^2}},$$

where S_b is the barrier position (whether S_u or S_d should be obvious from the example),

$$d_1 = \frac{\log(S/E) + \left(r - q + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = \frac{\log(S/E) + \left(r - q - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_3 = \frac{\log(S/S_b) + \left(r - q + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_4 = \frac{\log(S/S_b) + \left(r - q - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_5 = \frac{\log(S/S_b) - \left(r - q - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_6 = \frac{\log(S/S_b) - \left(r - q + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_7 = \frac{\log(SE/S_b^2) - \left(r - q - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_8 = \frac{\log(SE/S_b^2) - \left(r - q + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}.$$

Up-and-out call

$$Se^{-q(T-t)}(N(d_1) - N(d_3) - b(N(d_6) - N(d_8))) - Ee^{-r(T-t)}(N(d_2) - N(d_4) - a(N(d_5) - N(d_7))).$$

Up-and-in call

$$Se^{-q(T-t)}(N(d_3) + b(N(d_6) - N(d_8))) - Ee^{-r(T-t)}(N(d_4) + a(N(d_5) - N(d_7))).$$

Down-and-out call

1. $E > S_b$:

$$Se^{-q(T-t)}(N(d_1) - b(1 - N(d_8))) - Ee^{-r(T-t)}(N(d_2) - a(1 - N(d_7))).$$

2. $E < S_b$:

$$Se^{-q(T-t)}(N(d_3) - b(1 - N(d_6))) - Ee^{-r(T-t)}(N(d_4) - a(1 - N(d_5))).$$

Down-and-in call

1. $E > S_b$:

$$Se^{-q(T-t)}b(1 - N(d_8)) - Ee^{-r(T-t)}a(1 - N(d_7)).$$

2. $E < S_b$:

$$Se^{-q(T-t)}(N(d_1) - N(d_3) + b(1 - N(d_6))) - Ee^{-r(T-t)}(N(d_2) - N(d_4) + a(1 - N(d_5))).$$

Down-and-out put

$$-Se^{-q(T-t)}(N(d_3) - N(d_1) - b(N(d_8) - N(d_6))) + Ee^{-r(T-t)}(N(d_4) - N(d_2) - a(N(d_7) - N(d_5))).$$

Down-and-in put

$$-Se^{-q(T-t)}(1 - N(d_3) + b(N(d_8) - N(d_6))) + Ee^{-r(T-t)}(1 - N(d_4) + a(N(d_7) - N(d_5))).$$

Up-and-out put

1. $E > S_b$:

$$-Se^{-q(T-t)}(1 - N(d_3) - bN(d_6)) + Ee^{-r(T-t)}(1 - N(d_4) - aN(d_5)).$$

2. $E < S_b$:

$$-Se^{-q(T-t)}(1 - N(d_1) - bN(d_8)) + Ee^{-r(T-t)}(1 - N(d_2) - aN(d_7)).$$

Up-and-in put

1. $E > S_b$:

$$-Se^{-q(T-t)}(N(d_3) - N(d_1) + bN(d_6)) + Ee^{-r(T-t)}(N(d_4) - N(d_2) + aN(d_5)).$$

2. $E < S_b$:

$$-Se^{-q(T-t)}bN(d_8) + Ee^{-r(T-t)}aN(d_7).$$

The following charts show each of the above types of barrier option, as well as the underlying vanilla option.

Note that with out options the value of the barrier option ‘hugs’ the vanilla, except that it must be zero at the barrier. With in options, the barrier value hugs zero except that it becomes the vanilla value at the barrier.

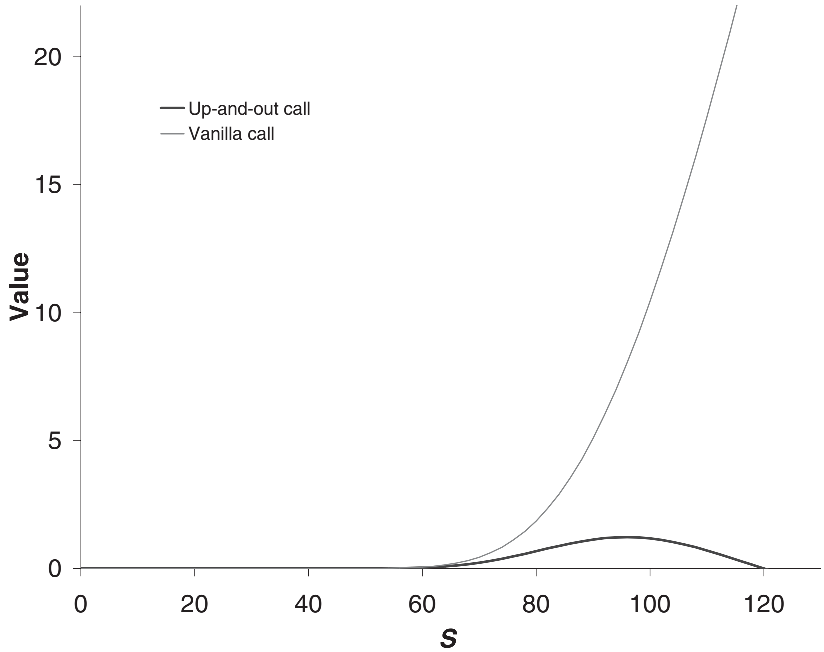

Figure 13.21 Up-and-out call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

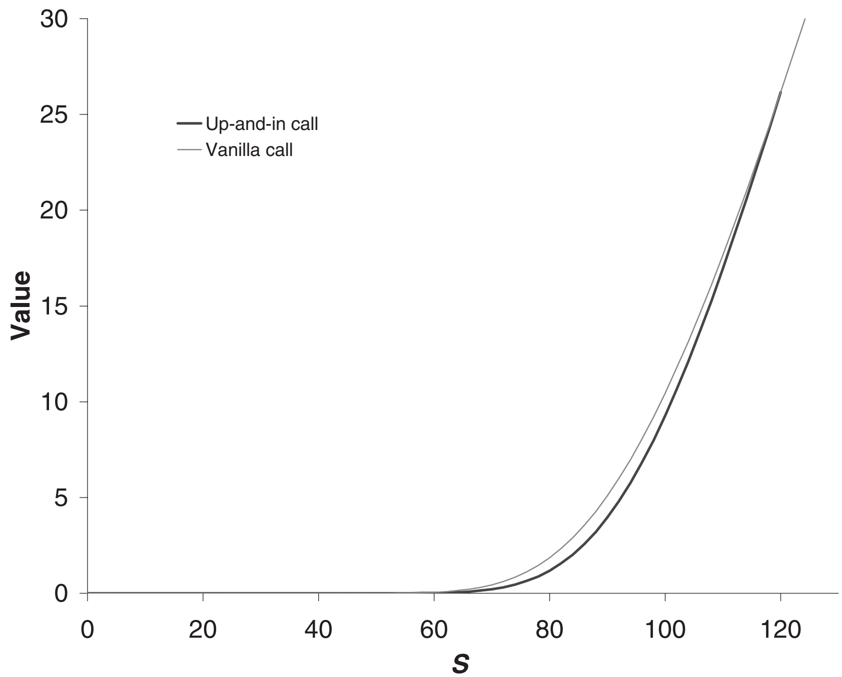

Figure 13.22 Up-and-in call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

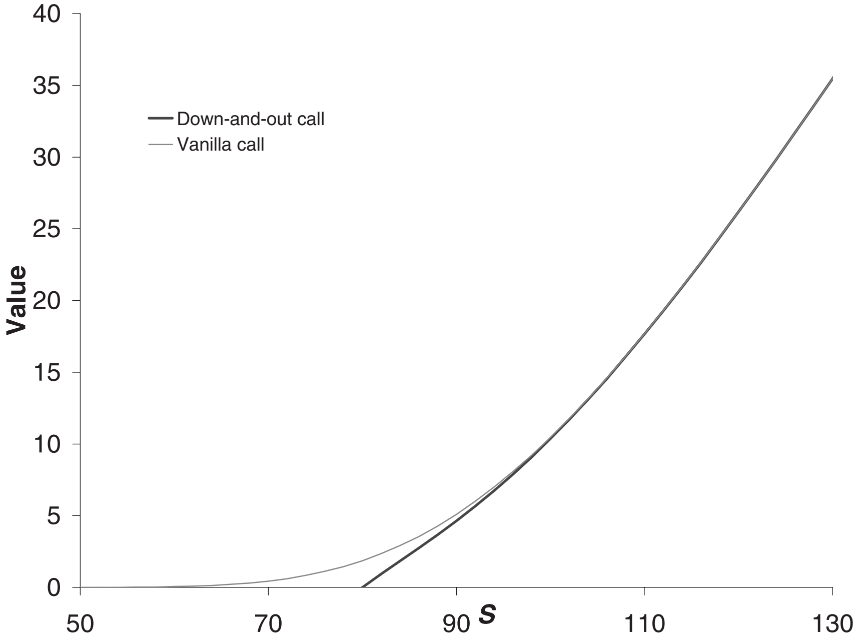

Figure 13.23 Down-and-out call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

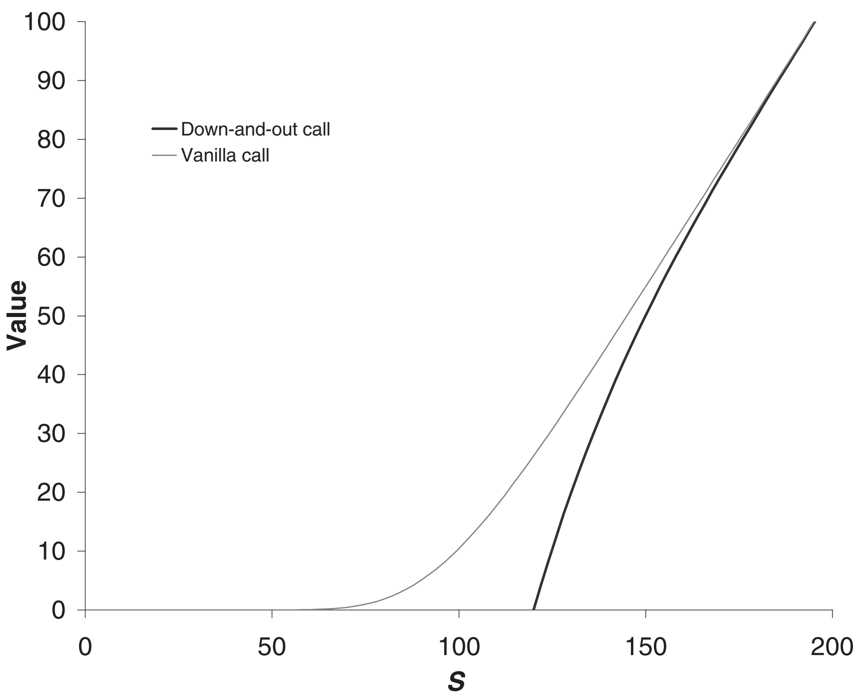

Figure 13.24 Down-and-out call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

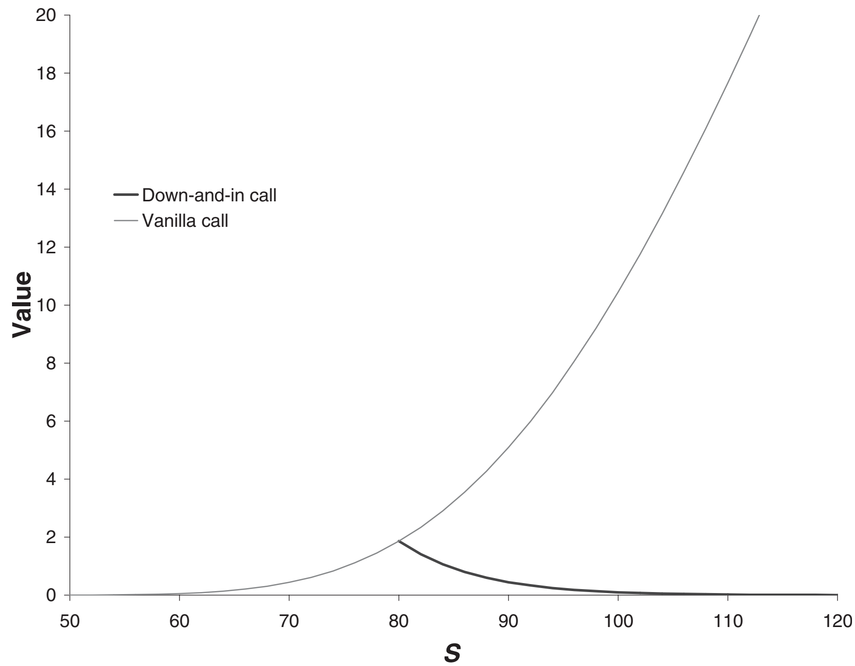

Figure 13.25 Down-and-in call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

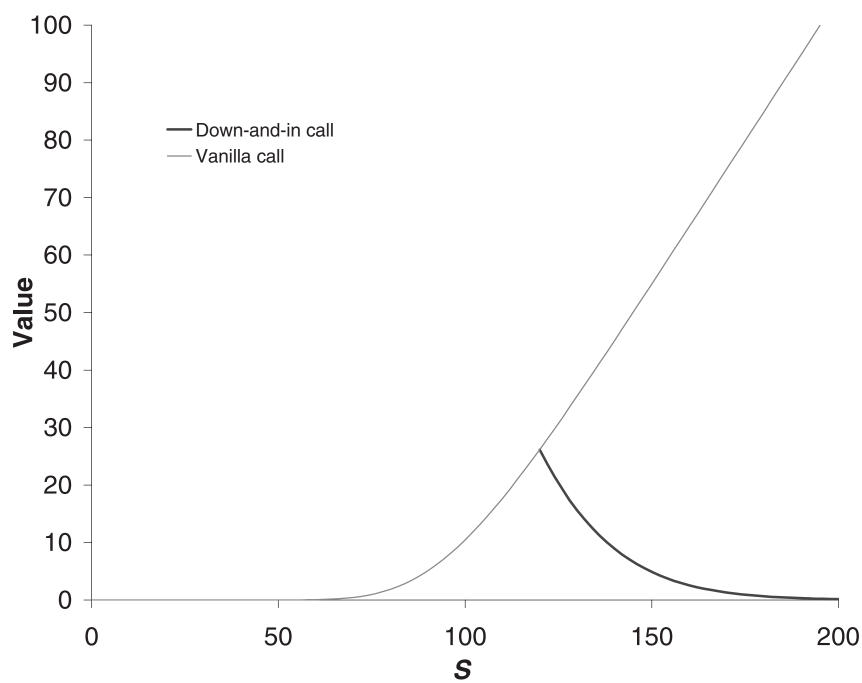

Figure 13.26 Down-and-in call. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

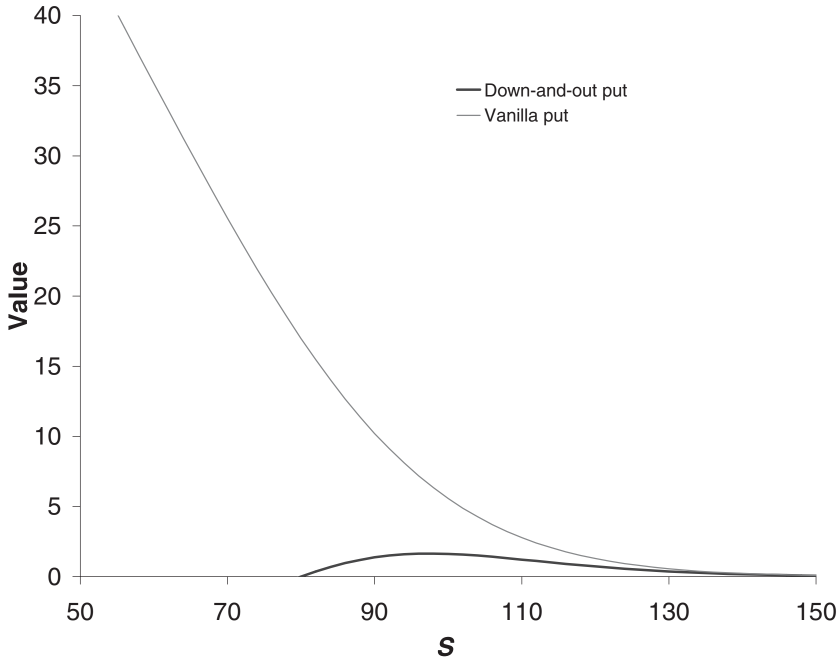

Figure 13.27 Down-and-out put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

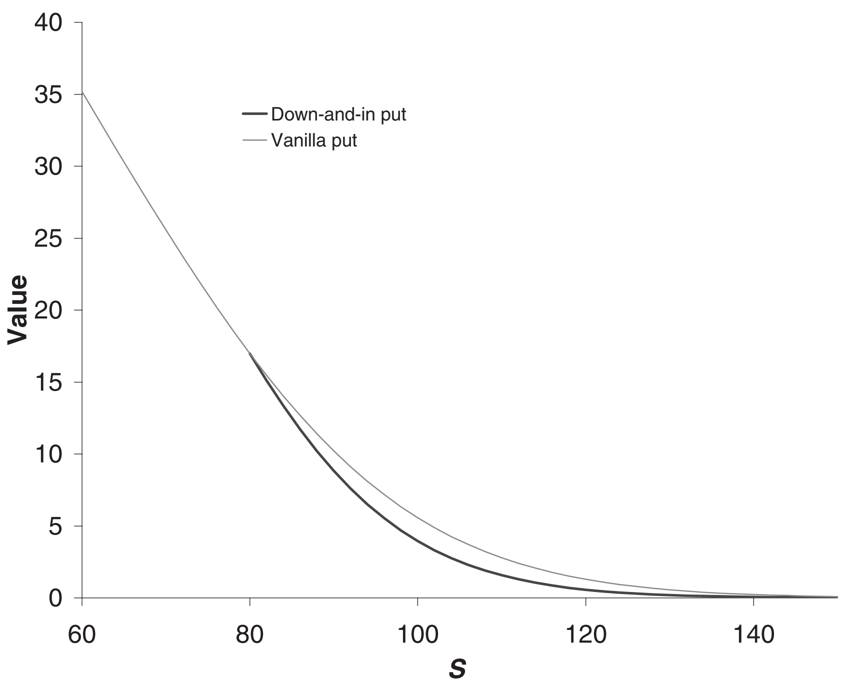

Figure 13.28 Down-and-in put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

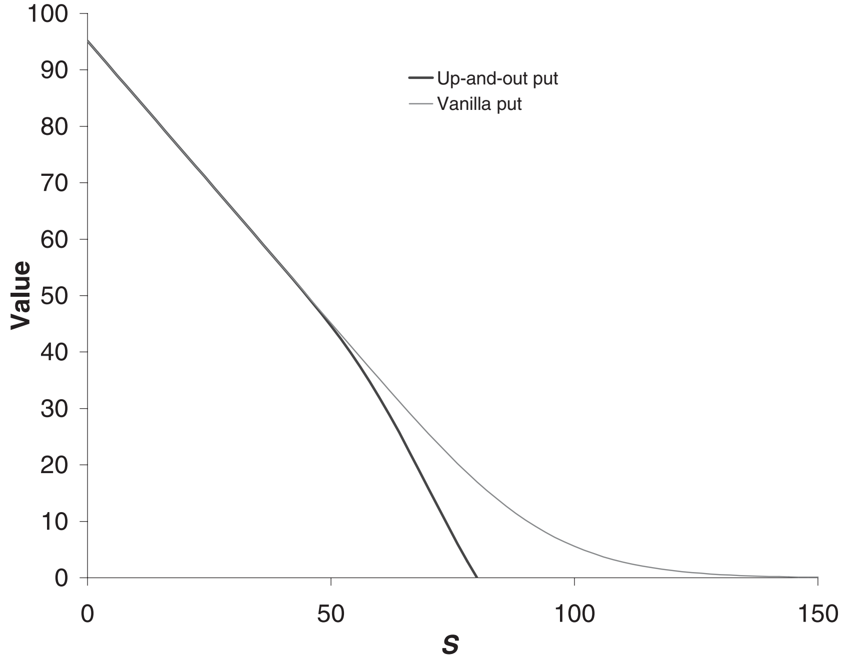

Figure 13.29 Up-and-out put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

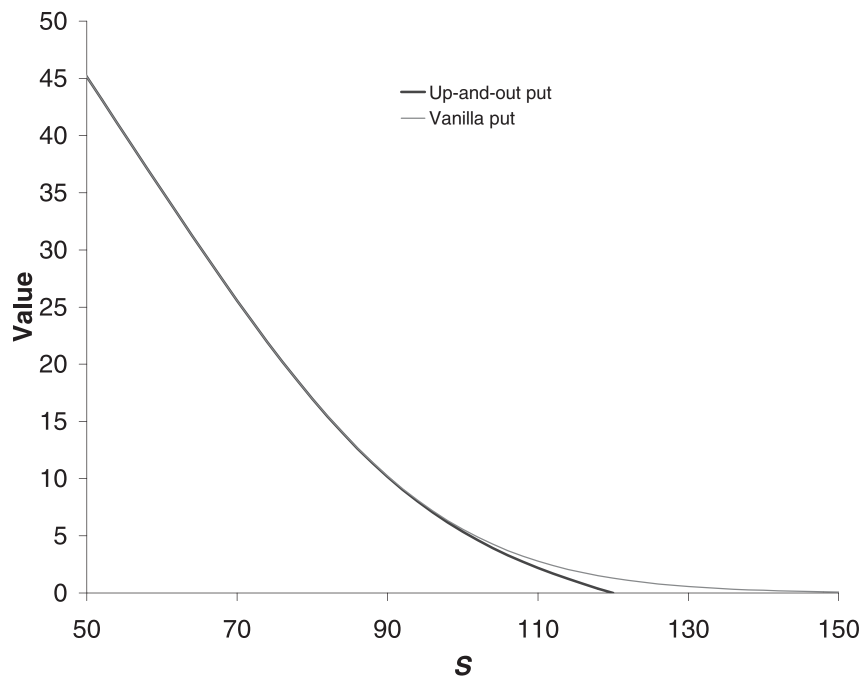

Figure 13.30 Up-and-out put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

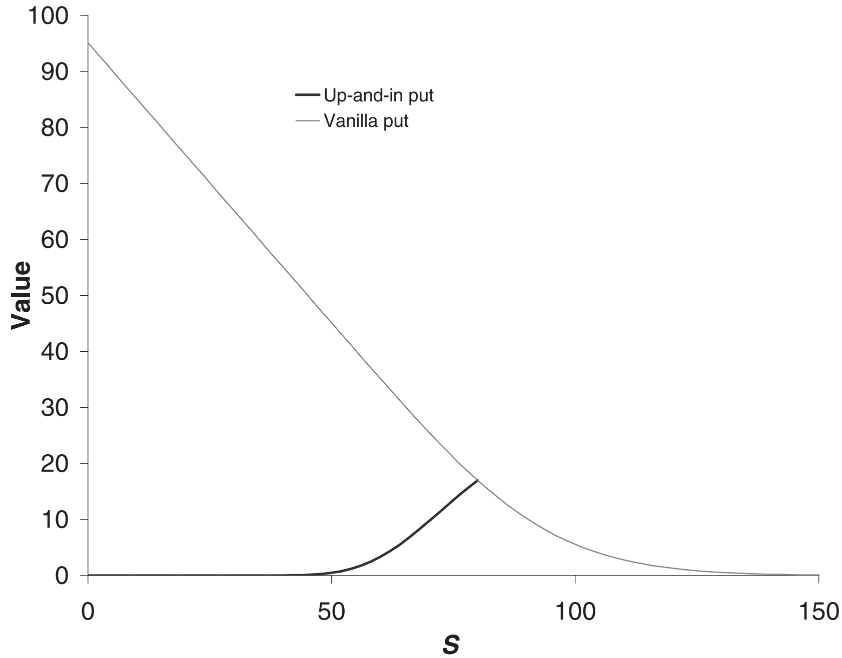

Figure 13.31 Up-and-in put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 80$.

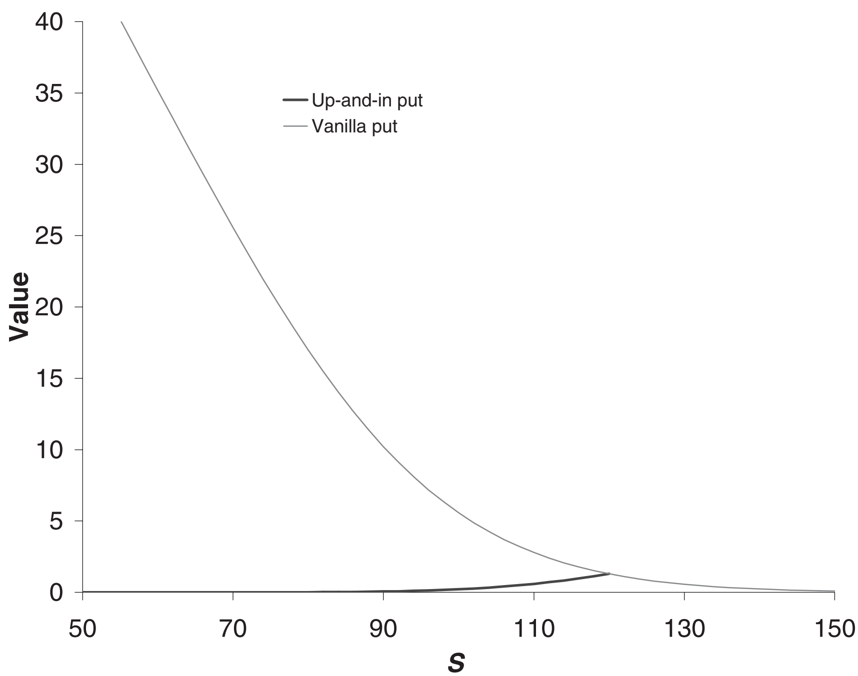

Figure 13.32 Up-and-in put. $\sigma = 0.2$, $r = 0.05$, $q = 0$, $E = 100$, $T = 1$ and $S_b = 120$.

FURTHER READING

- Many of the original barrier formulæ are due to Reiner & Rubinstein (1991).
- The formulæ above are explained in Taleb (1997) and Haug (2007). Taleb discusses barrier options in great detail, including the reality of hedging that I have only touched upon.
- The article by Carr (1995) contains an extensive literature review as well as a detailed discussion of protected barrier options and rainbow barrier options.
- See Derman *et al*. (1997) for a full description of the static replication of barrier options with vanilla options.
- See Carr (1994) for more details of put-call symmetry.
- See Haug & Haug (2002) for the pricing of barrier options that depend on two underlying assets.
- More closed-form solutions can be found in Banerjee (2003).

Time Out...

Binomial model revisited

Using trees to price barrier options can be a bit of a chore. This is because it is tedious to line up nodes with the barrier level. You can be faced with the question of on which node to set the option value to zero. Interpolation methods make the most sense. Those fond of trees will go to extraordinary lengths to justify various *ad hoc* tree modifications. In the finite-difference method the barrier is trivial — the mathematician's favorite word — to incorporate.

EXERCISES

1. Check that the solution for the down-and-out call option, $V_{D/O}$, satisfies Black–Scholes, where

$$V_{D/O}(S,t) = C(S,t) - \left(\frac{S}{S_d}\right)^{1-\frac{2r}{\sigma^2}} C(S_d^2/S, t),$$

and $C(S,t)$ is the value of a vanilla call option with the same maturity and payoff as the barrier option.

Hint: Show that $S^{1-\frac{2r}{\sigma^2}}V(X^2/S,t)$ satisfies Black–Scholes for any X, when $V(S,t)$ satisfies Black–Scholes.

2. Why do we need the condition $S_d < E$ to be able to value a down-and-out call by adding together known solutions of the Black–Scholes equation (as in question 1)? How would we value the option in the case that $S_d > E$?

3. Check the value for the down-and-in call option using the explicit solutions for the down-and-out call and the vanilla call option.

4. Formulate the following problem for the accrual barrier option as a Black–Scholes partial differential equation with appropriate final and boundary conditions:

   The option has barriers at levels S_u and S_d, above and below the initial asset price, respectively. If the asset touches either barrier before expiry then the option knocks out with an immediate payoff of $\Phi(T-t)$. Otherwise, at expiry the option has a payoff of $\max(S-E,0)$.

5. Formulate the following barrier option pricing problems as partial differential equations with suitable boundary and final conditions:
   (a) The option has barriers at levels S_u and S_d, above and below the initial asset price, respectively. If the asset touches both barriers before expiry, then the option has payoff $\max(S-E,0)$. Otherwise the option does not pay out.
   (b) The option has barriers at levels S_u and S_d, above and below the initial asset price, respectively. If the asset price first rises to S_u and then falls to S_d before expiry, then the option pays out \$1 at expiry.

6. Price the following double knockout option: the option has barriers at levels S_u and S_d, above and below the initial asset price, respectively. The option has payoff \$1, unless the asset touches either barrier before expiry, in which case the option knocks out and has no payoff.

7. Prove put-call parity for simple barrier options:

$$C_{D/O}+C_{D/I}-P_{D/O}-P_{D/I}=S-Ee^{-r(T-t)},$$

   where $C_{D/O}$ is a European down-and-out call, $C_{D/I}$ is a European down-and-in call, $P_{D/O}$ is a European down-and-out put and $P_{D/I}$ is a European down-and-in put, all with expiry at time T and exercise price E.

8. Why might we prefer to treat a European up-and-out call option as a portfolio of a vanilla European call option and a European up-and-in call option?